Unit 1 – Using Objects and Methods Name: **KEY**
HW 4

1) Complete the following table by predicting the output of each statement.

	Code	Expected Output
a)	`System.out.println(7 / 2);`	3
b)	`System.out.println(7.0 / 2);`	3.5
c)	`System.out.println((double) 7 / 2);`	3.5
d)	`System.out.println((double) (7 / 2));`	3.0

2) Explain, in your own words, why there is a difference between `(double) 7 / 2` and `(double) (7 / 2)`.

In (double) 7/2, the 7 is casted to a double first, then the division is performed. Since the division is between a double and int, the result is a double resulting in 3.5. In (double)(7/2), the division is performed first and since it is between two ints, integer division is performed resulting in an int of 3. Then the cast is performed, turning the 3 into the double of 3.0.

3) The following code contains an error.

```
int x = (double) 7 / 2;
```

a) What type of error is this? Syntax error

b) Why is this an issue? You are trying to fit a double value into an int, and there is not enough space reserved for it.

c) How can you fix this issue? Change "int x" to "double x" in the declaration.

4) Complete the following table by predicting the output of each statement.

	Code	Expected Output

a)	`double value = 5.7;`
	`int x = (int) (value + 0.5);`
	`System.out.println(x);`
		6

b)	`double value = 2.3;`
	`int y = (int) (value + 0.5);`
	`System.out.println(y);`
		2

c)	`double value = 8.9;`
	`int z = (int) (value + 0.5);`
	`System.out.println(z);`
		9

5) Imagine you are writing a grading program that needs to round students' average scores.

a) Why would `int roundedScore = (int) (score + 0.5);` be a useful trick?

We do not know which direction we would want to round, so this trick makes sure that we round in the correct direction every time.

b) What would happen if we just used `(int) score;` instead?

This would truncate the score every time, so even if your score is 89.9, it would "round" it to a 89 when it should really be rounded to a 90.

6) Consider the code segment below.

```
int num = 5/2;
```

Circle or highlight the statements that are equivalent to the code segment above.

I. `int num = (int)(5/2);`

II. `int num = (int)(5.0/2);`

III. `int num = 5.0/2.0;`

IV. `int num = (int) 5/2;`

V. `int num = (double) 5 / (double) 2;`

7) Determine the final value of the expressions below.

a) `8 / 3 + (double) 5 / 2`

```
2 + 2.5
4.5
```

b) `(double) 10 / 4 + 3 / 2`

```
2.5 + 1
3.5
```

c) `10.5/2 + (int) 4.9 + 3.25`

```
5.25 + 4 + 3.25
12.5
```

d) `20.0/4 + (double) (15/2) - (int) 5.5`

```
5.0 + 7.0 - 5
7.0
```